HEADWAY VOICE · STORYBOARD IN CHROME · SAUTI ADAPT TEAM

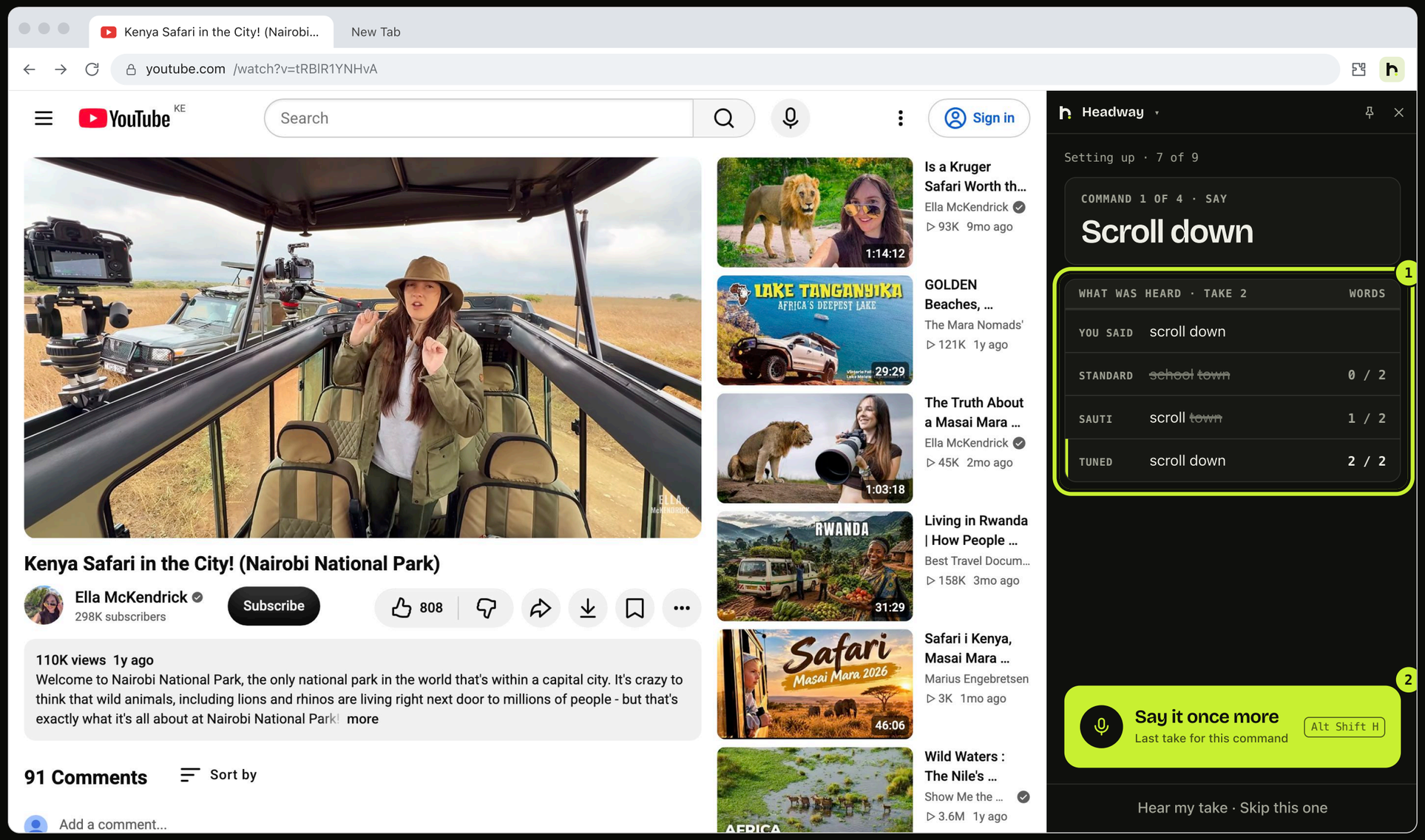
One voice, real pages, step by step.

Headway's side panel on real YouTube pages, as Faridah would use it: set up once, press her key, speak. Common commands go straight from the Sauti transcript to Chrome. Anything else goes through Gemini, which only chooses from actions the page really has. The pages are real captures (signed out, ads removed). The side panel is the design laid over them, not a working build yet. Times and transcripts are sample values.

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01 Set up once → 02 Press her key → 03 Command list → 04 Gemini lane → 05 Done → 06 Not sure? Ask

01 **Set up once, in her own voice** she reads “Scroll down” three times

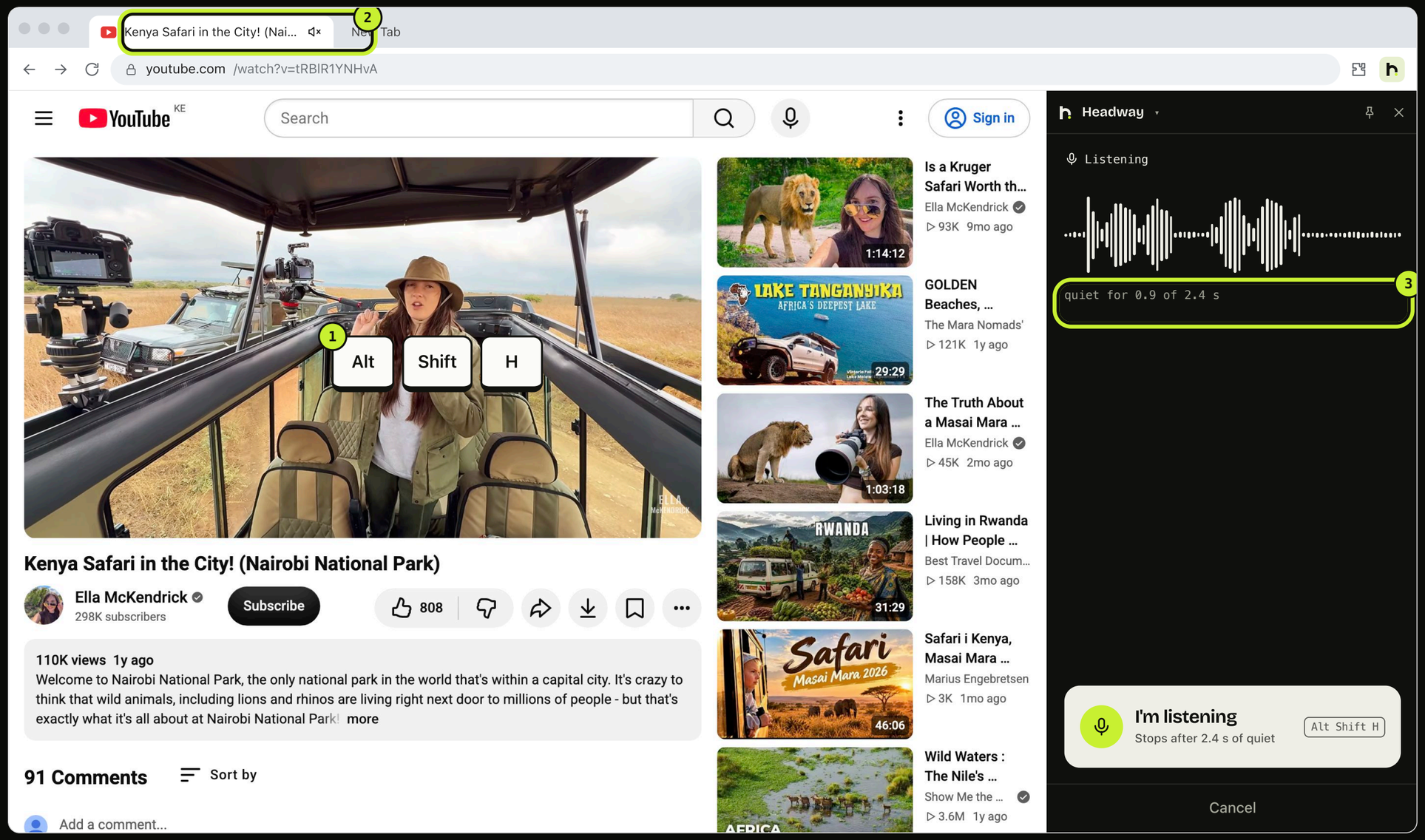


WHAT TO NOTICE

1 **Three readings of one take**
Chrome's standard recognition, the Sauti model as shipped, and Sauti with her calibration. This is the difference the project makes, shown in her own words.

2 **Saved before it's scored**
Every take is kept locally first, so she never repeats one. Take 3 is held back to score her honestly at the end.

02 # She presses her key, from the page no clicking into the panel



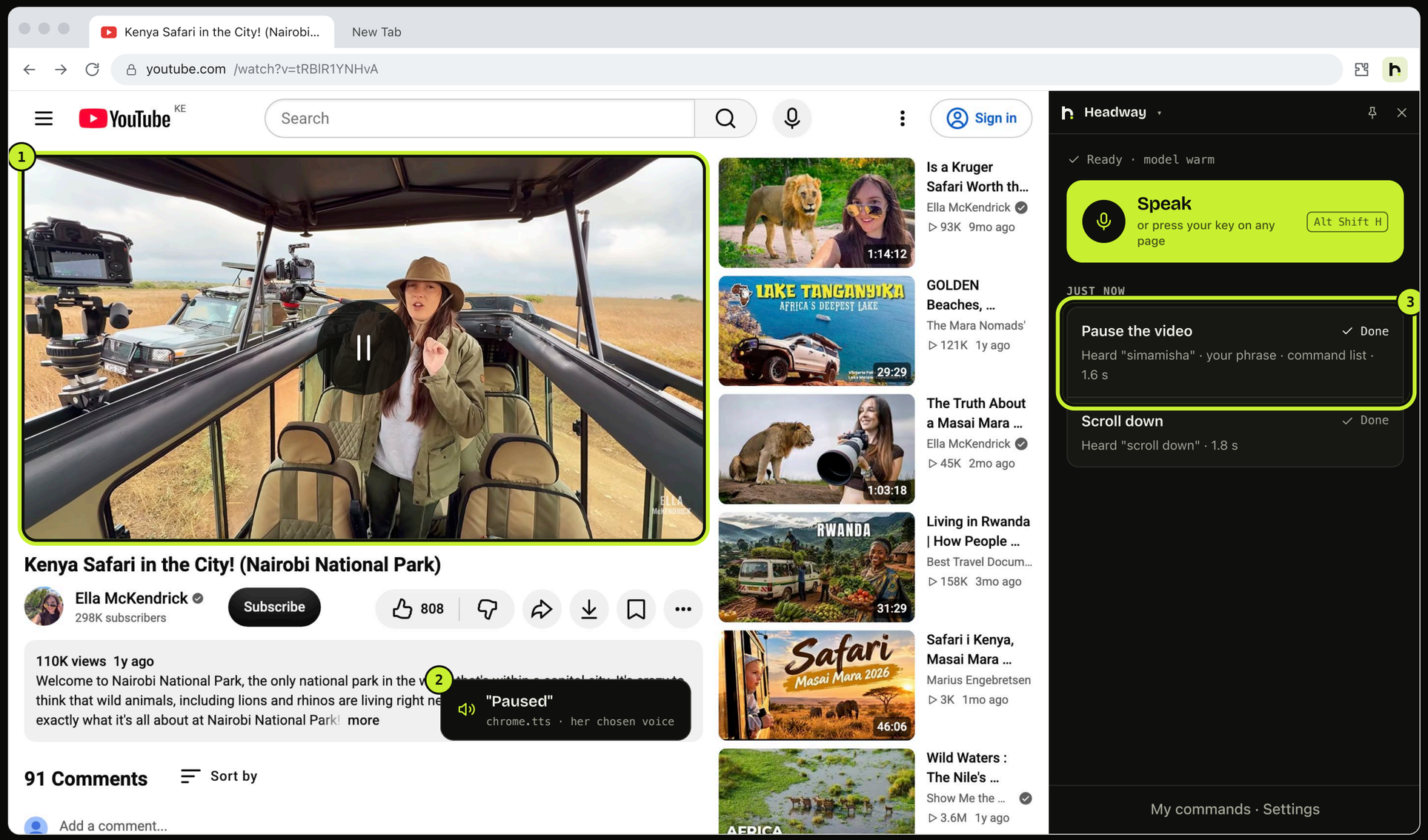
WHAT TO NOTICE

1. **Her key works on any page**
   `chrome.commands` gives one shortcut that fires even while YouTube has focus. She chose it in onboarding; a switch or the big panel button work the same way.

2. **The tab goes quiet**
   The video's sound is muted while she speaks (`chrome.tabs.update({muted})`), so the model hears her, not the safari.

3. **Her pace, not Chrome's**
   It waits 2.4 s of quiet before it stops, the value set from her own pauses in onboarding.

03 Fast path: the command list she says “simamisha” (her phrase for pause)



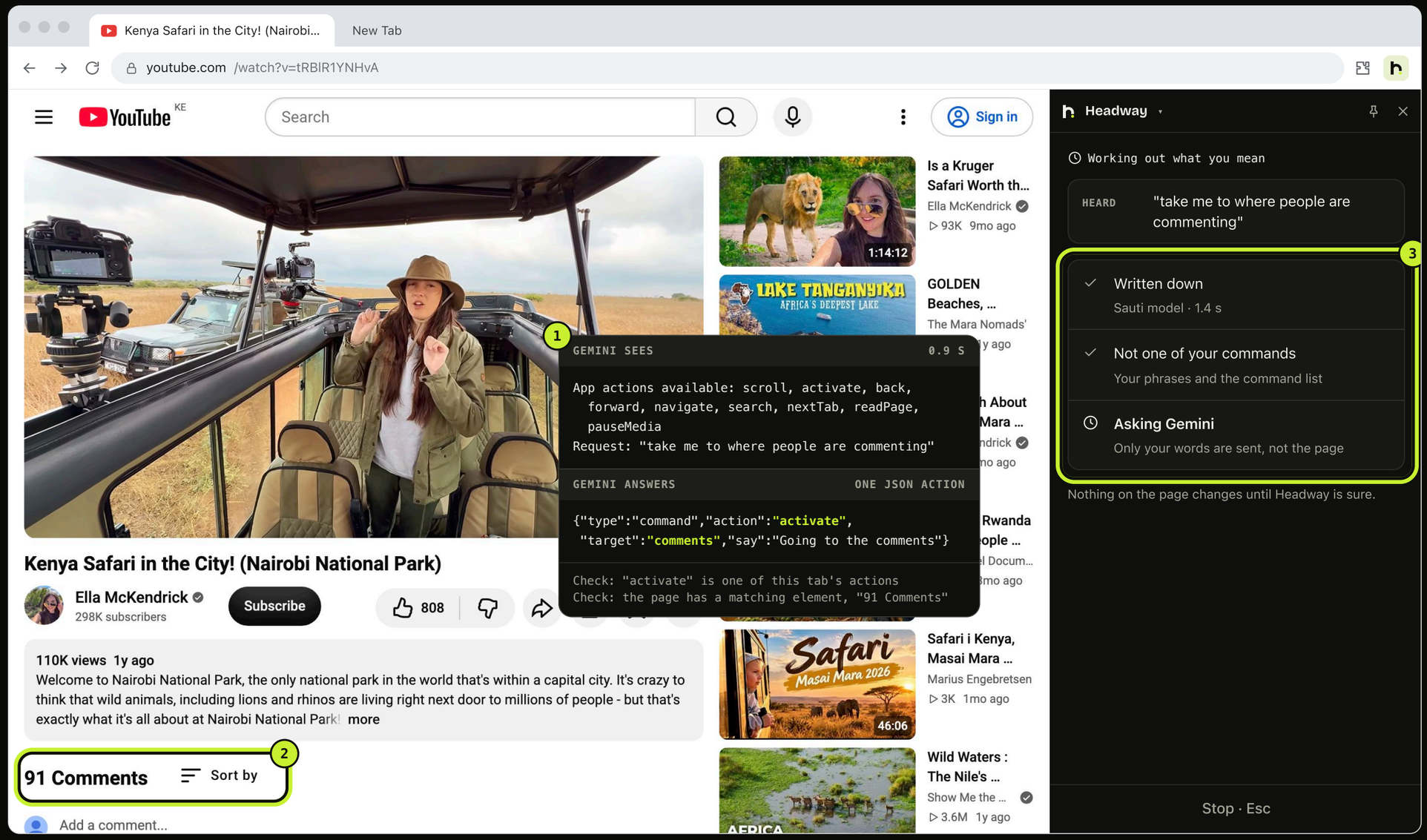
WHAT TO NOTICE

1. **No Gemini call**
   "simamisha" is her own alias for pause, matched on this computer before the toolkit grammar even runs. Most commands take this path.

2. **Feedback the way she chose**
   Spoken, a sound, or text only. Set in onboarding, not guessed.

3. **Timings are visible**
   Every history row says which path ran and how long it took, so the team can measure real latency.

04 Gemini lane: when the list doesn't know it she says “take me to where people are commenting”



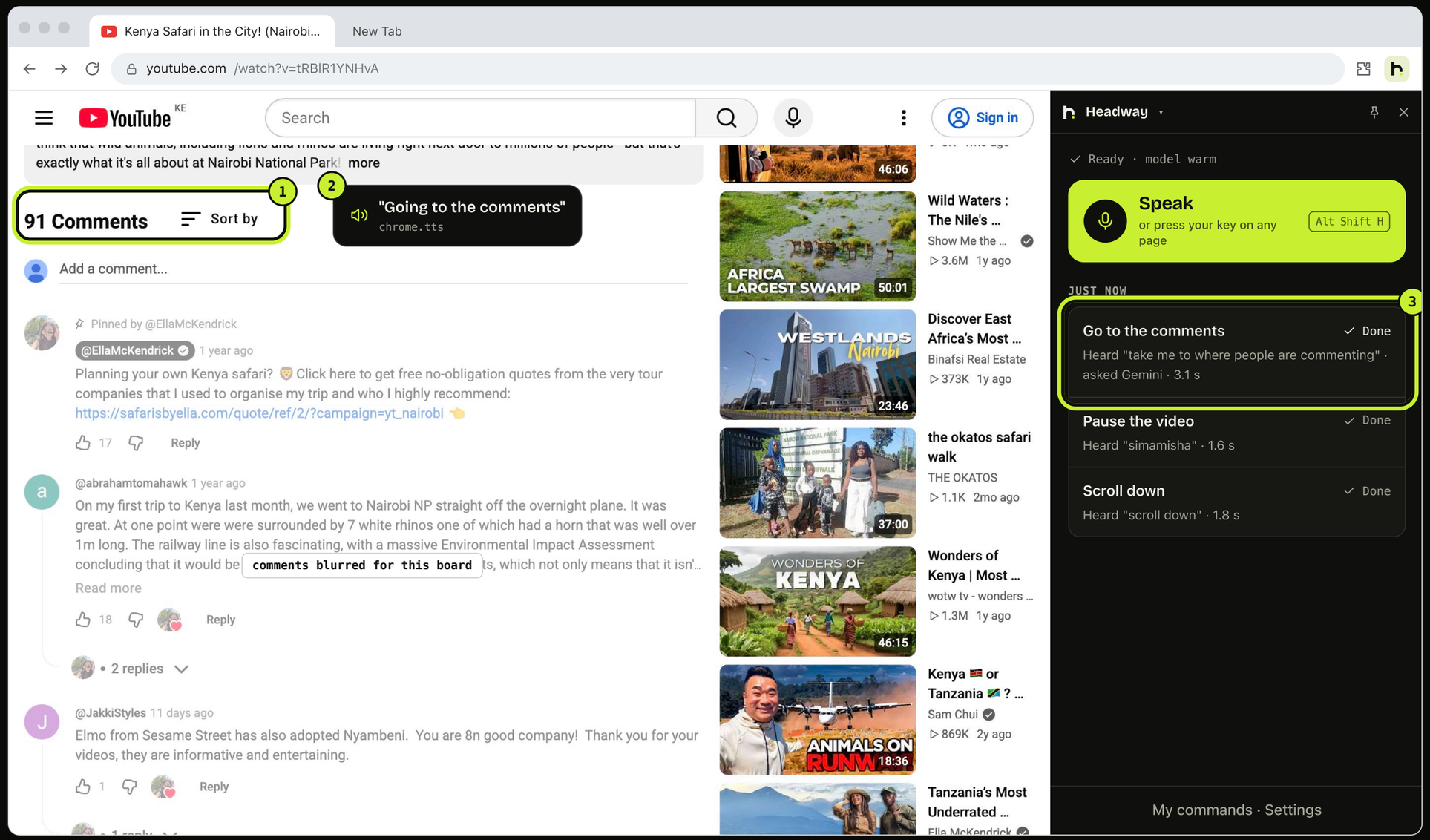
WHAT TO NOTICE

1. **Gemini gets her words and a menu**
   The toolkit's `llm-lane.js` sends only the transcript and the actions this tab supports. No page text, so a web page can't instruct Gemini. Nano on the device first, cloud Gemini through our relay as a fallback.

2. **Checked against the real page**
   The Chrome port looks for a visible element matching "comments" and finds the "91 Comments" heading. If nothing matches, she gets "Did you mean..." (scene 06).

3. **She can see the slow path**
   Each step is listed, with Stop on Esc (the Controller's `stop()`).

05 # Chrome acts, Headway answers 3.1 s from her last word



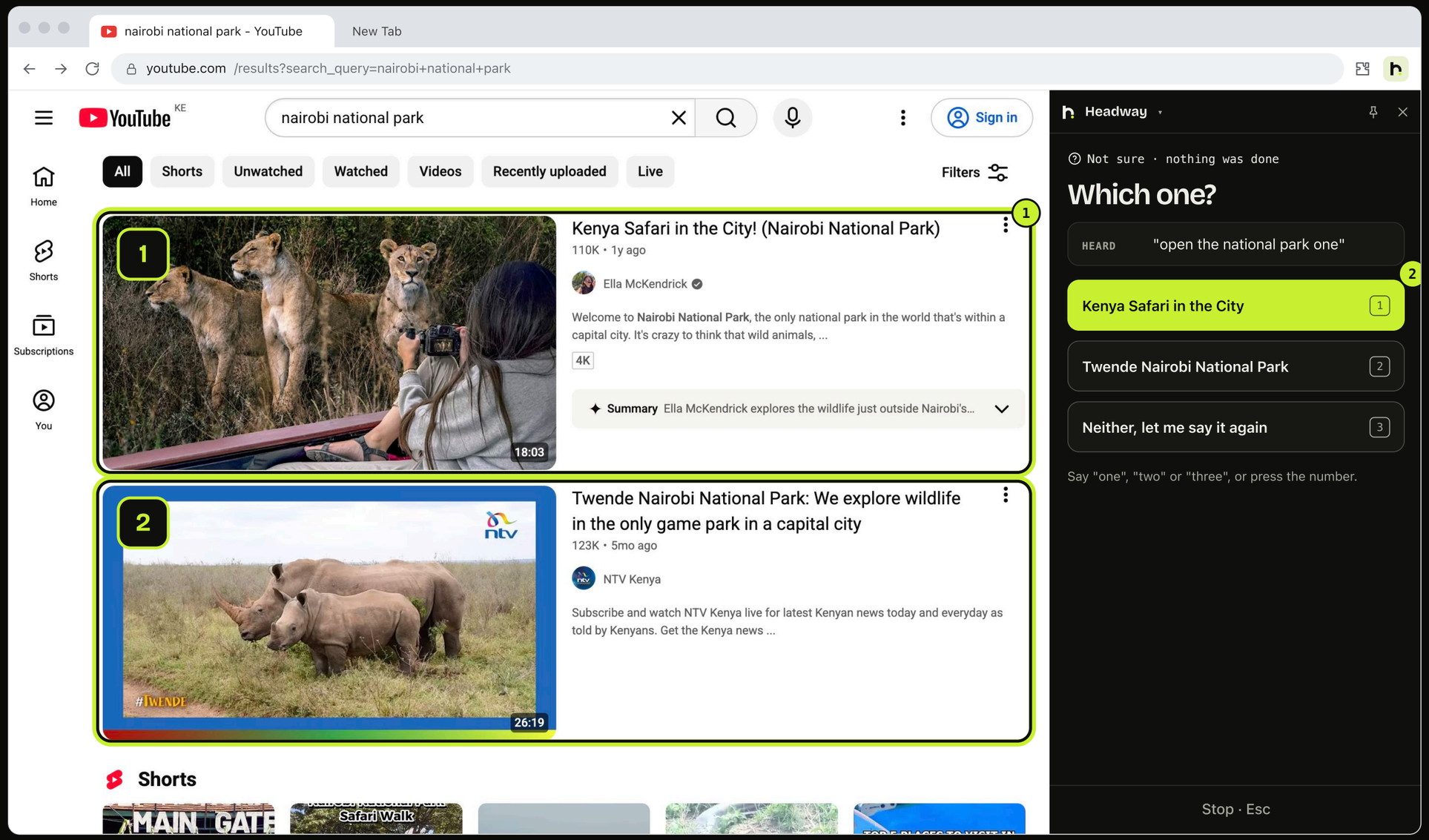
WHAT TO NOTICE

1 The action happened on the page
The content script scrolled to the "91 Comments" heading and moved focus there, so the next "read this" starts in the right place.

2 She hears it
Gemini's short "say" phrase is spoken back, which also confirms what Headway understood.

3 The history keeps score
"asked Gemini · 3.1 s" against 1.6 s for the command list: the cost of the slow path, in the open.

06 # Not sure? Nothing runs. She chooses. she says “open the national park one”



WHAT TO NOTICE

1. **Numbers on the page match the panel**
   Both videos say "national park", so Headway labels both on the page and asks. Default to asking: unclear speech never clicks anything.

2. **Big, few, numbered choices**
   She can answer by voice ("one"), a key, or a tap. Her answer can become a remembered phrase, as a toolkit proposal she accepts.

Pages captured from youtube.com on 2026-11-07 in a signed-out Chrome profile; ads and sign-in prompts removed, comments blurred. The Headway panel is the design from `voice-onboarding.html`, overlaid; the extension is not built yet. All transcripts, times and scores are samples. Plan: `apps/headway/docs/VOICE_PLAN.md`.

HEADWAY VOICE · DESIGN BOARD · SIDE PANEL 380 × 760

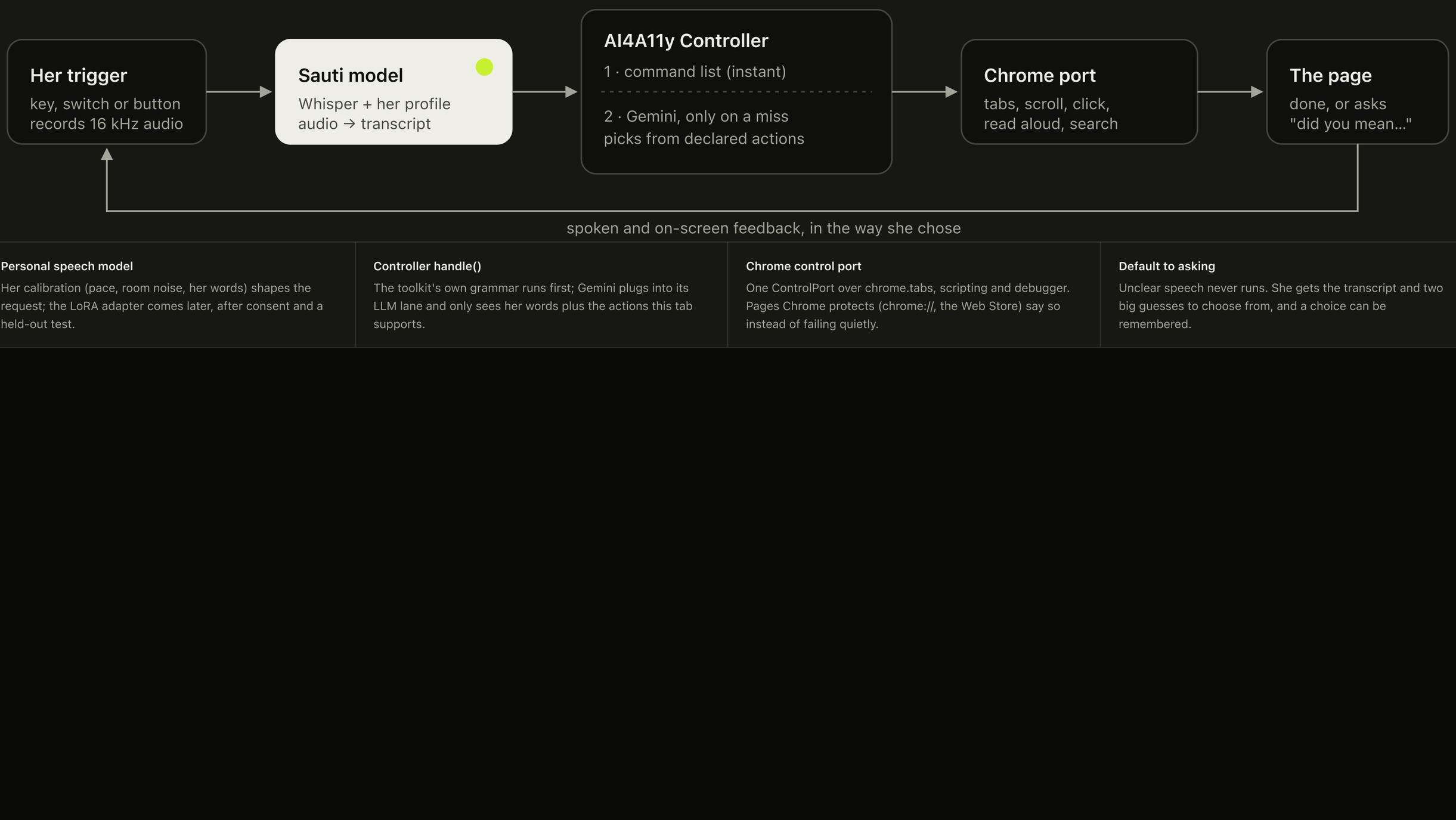
Teach Headway your voice.

The audio-only Headway: a person speaks, a speech model tuned to them writes it down, the AI4A11y Controller works out what they mean, and Chrome does it on the page. Onboarding is where the tuning happens, and where the person sees, in their own recordings, why it matters. Sentences, scores and transcripts on this board are sample values, not results.

How one command travels

whiteboard → system

FARIDAH SPEAKS → THE BROWSER ACTS



Personal speech model

Her calibration (pace, room noise, her words) shapes the request; the LoRA adapter comes later, after consent and a held-out test.

Controller handle()

The toolkit's own grammar runs first; Gemini plugs into its LLM lane and only sees her words plus the actions this tab supports.

Chrome control port

One ControlPort over chrome.tabs, scripting and debugger. Pages Chrome protects (chrome://, the Web Store) say so instead of failing quietly.

Default to asking

Unclear speech never runs. She gets the transcript and two big guesses to choose from, and a choice can be remembered.

When the command list misses: the Gemini lane `controller/llm-lane.js`

EXAMPLE: SHE SAYS "TAKE ME TO WHERE PEOPLE ARE COMMENTING"

1 Sauti writes it down — ✓ 1.4 s

Her audio goes to the Sauti model on Modal.

```
"take me to where people are commenting"
```

2 Her aliases, then the toolkit grammar — – Miss

Instant, on this computer. No pattern matches "take me to … commenting".

```
grammar.parse()  →  miss
```

3 Ask Gemini, with only her words and this tab's actions — 0.9 s

Gemini Nano on the device first, cloud Gemini through our relay if Nano is missing. Page text is never in the prompt, so a web page can't give Gemini instructions.

```
Settings you may change: fontScale (50–300), darkMode, …
App actions available: scroll, activate, back, forward,
  navigate, search, nextTab, previousTab, readPage, pauseMedia
Request: "take me to where people are commenting"
JSON:
```

4 Gemini answers with one JSON action — ✓ Parsed

```
{"type":"command","action":"activate",
 "target":"comments","say":"Going to the comments"}
```

5 Checked before anything runs — ✓ Allowed

Is "activate" one of this tab's actions? Yes. Does a visible element match "comments"? The Chrome port finds *Comments · 1,204*. Anything else (unknown action, invented setting, no match) becomes "Did you mean…" and nothing runs.

6 Her confirmation rule — ✓ No check needed

She chose "ask before sending forms". This is a link, so it runs. A form would stop at "Send this form?".

7 Chrome acts, Headway answers — ✓ Done

The content script scrolls to and focuses the comments; `chrome.tts` says "Going to the comments". The history row shows "asked Gemini · 3.1 s".

Onboarding: calibrate to one voice

9 steps · about 12 minutes · saves after every take

01 · Welcome — first open

Headway 1 / 9

SET UP YOUR VOICE

Use the web by speaking, the way you speak.

You'll say a few short commands so Headway learns your voice. Everything you say is saved as you go, so you never have to repeat a take.

- About 12 minutes
  Stop whenever you like and carry on later.
- Rest when you need to
  A break comes up every few commands.
- Kenyan English or Kiswahili
  Or both. You choose your own commands.

Start Enter

Someone is helping me set up

Why the second button: many co-designers set up with a family member or therapist. It switches the copy to "they" and keeps her as the one who speaks. It never asks about a diagnosis.

02 · Your recordings — consent

Headway 2 / 9

Where your voice goes

Headway only listens when you start it.

- Recordings stay on this computer
  In Headway's storage. Not in the cloud, not in a chat.
- Each command is sent to write it down
  To the Sauti speech server, then the audio is not kept there.
- Delete everything with one button
  In Settings, any time.

- [x] I agree to Headway recording my voice for this setup.
- [ ] Also use my recordings to train a model just for me.
  Optional. You can say yes later.

I agree

Not now

Two separate yeses. Using the extension and training a personal adapter are different permissions (AGENTS.md rules 4–5). "Not stored on the server" must be true of the Modal endpoint before this copy ships.

03 · Microphone and room — noise floor

Headway 3 / 9

Stay quiet for five seconds

Headway is listening to the room, so it can tell your voice from the background.

ROOM SOUND

✓ Quiet room — MacBook microphone ▾

NOW SAY YOUR NAME, OR ANY WORD

✓ Heard you clearly 4.6 s

Next

Use a different microphone

Sets the silence floor the server already uses (MIN_RMS), per person and per room, instead of one fixed number. The first take is never scored.

04 · How you start talking — her trigger

Headway 4 / 9

How would you like to start?

Pick what's easiest. You can change it later.

- (•) One key, from any page
  Alt + Shift + H. Press again to finish.
- ( ) A big button in this panel
  Tap to start, tap to finish.
- ( ) My switch
  Press your switch once; we'll learn which key it sends.

FINISHING

- (•) When I stop talking
  Headway learns your pauses next.
- ( ) When I press again
  No time limit while you speak.

Try it Alt Shift H

chrome.commands gives a key that works on any page, even when the page has focus. Stored with the toolkit's `shortcut` block, so it follows her profile. No wake word: nothing listens in the background.

05 · Your pace — end-of-speech wait

Headway 5 / 9

Take your time. We'll wait.

You said "Open the first video on YouTube". Here is where a standard listener stops, and where Headway does.

STANDARD LISTENING — *"video on YouTube" lost* — stops after 0.8 s quiet

HEADWAY, YOUR PACE — 1.9 s pause kept — waits 2.4 s, stops

WAIT AFTER YOU STOP 2.4 s

1 s — 5 s

Keep 2.4 seconds

Say it again

The first "aha". Same audio, two cut-off rules, drawn from her own recording. The suggested wait is her longest mid-sentence pause plus a margin; she can move it.

06 · Your words — her command set

Headway 6 / 9

Which commands do you want?

Keep the ones you'll use. Say them however you like.

English | Kiswahili | Both

- Scroll down — also: "shuka chini"
- Go back — also: "rudi nyuma"
- Read this to me — also: "nisomee"
- Next tab — no Kiswahili phrase yet

+ Add my own phrase

Record these · 4 commands

She chooses, we don't infer. Each phrase maps to one action the Chrome port declares. Her own phrase ("nisomee") becomes an alias, stored in her toolkit profile.

07 · Record a command — take 2 of 3

Headway 7 / 9

COMMAND 1 OF 4 · SAY

Scroll down

● ● ○ 2 saved

WHAT WAS HEARD · TAKE 2		WORDS
YOU SAID	scroll down	
STANDARD	~~school~~ ~~town~~	0 / 2
SAUTI	scroll ~~town~~	1 / 2
TUNED	scroll down	2 / 2

Struck-through words were heard wrong.

Say it once more — Last take for this command — Alt Shift H

▷ Hear my take — Skip this one

The showcase, take by take. One recording, three readings: Chrome's standard recognition, the Sauti model as it ships, and Sauti with her calibration. Takes 1–2 teach Headway

07 · Listening — state

Headway 7 / 9

COMMAND 1 OF 4 · SAY

Scroll down

● ● ○ take 3

Listening — quiet for 0.9 of 2.4 s

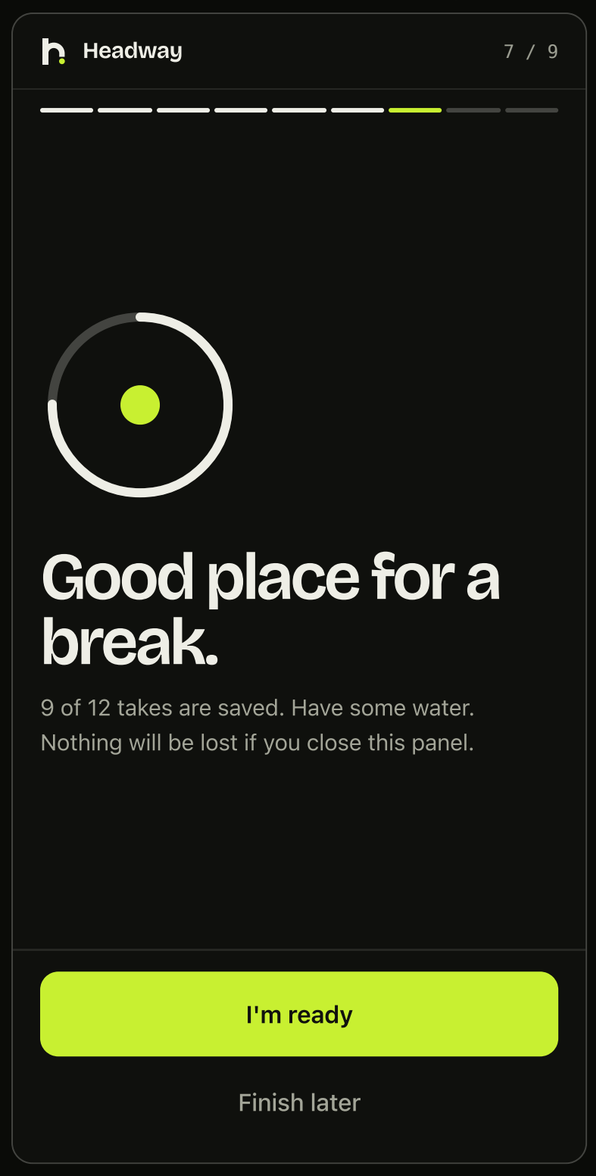
I'm listening
Stops after 2.4 s of quiet, or press to finish — Alt Shift H

Cancel this take

The countdown is visible, so a long pause never feels like a mistake. Media in the tab is muted while she speaks (the Controller's `muteAudio`).

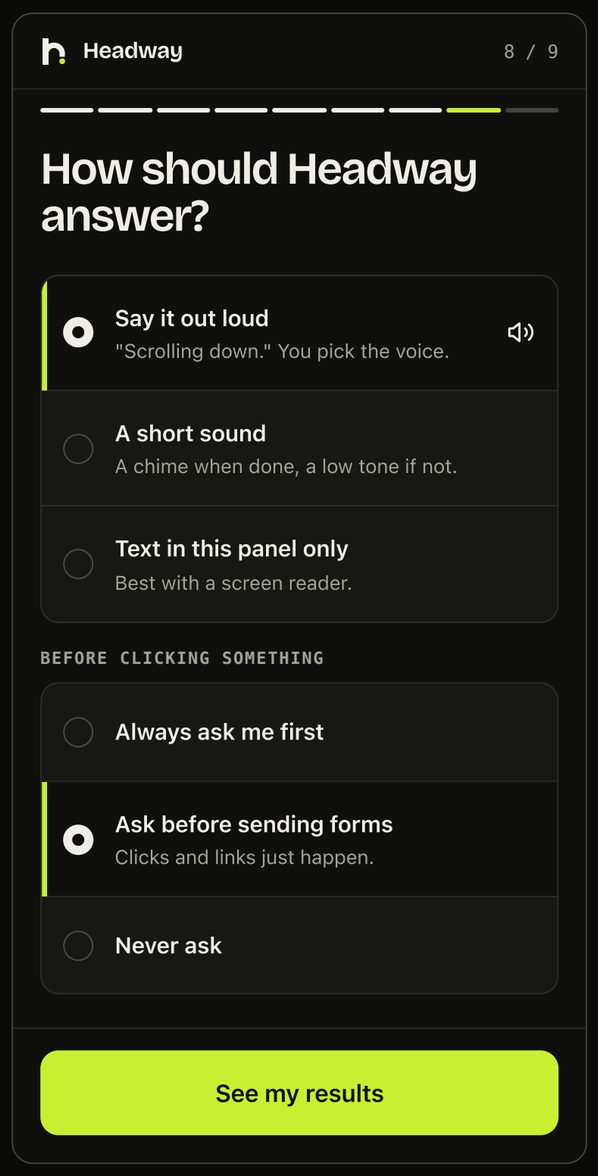
(aliases, word hints); take 3 is held back and only used to score her in step 9.

08 · Rest every 3 commands



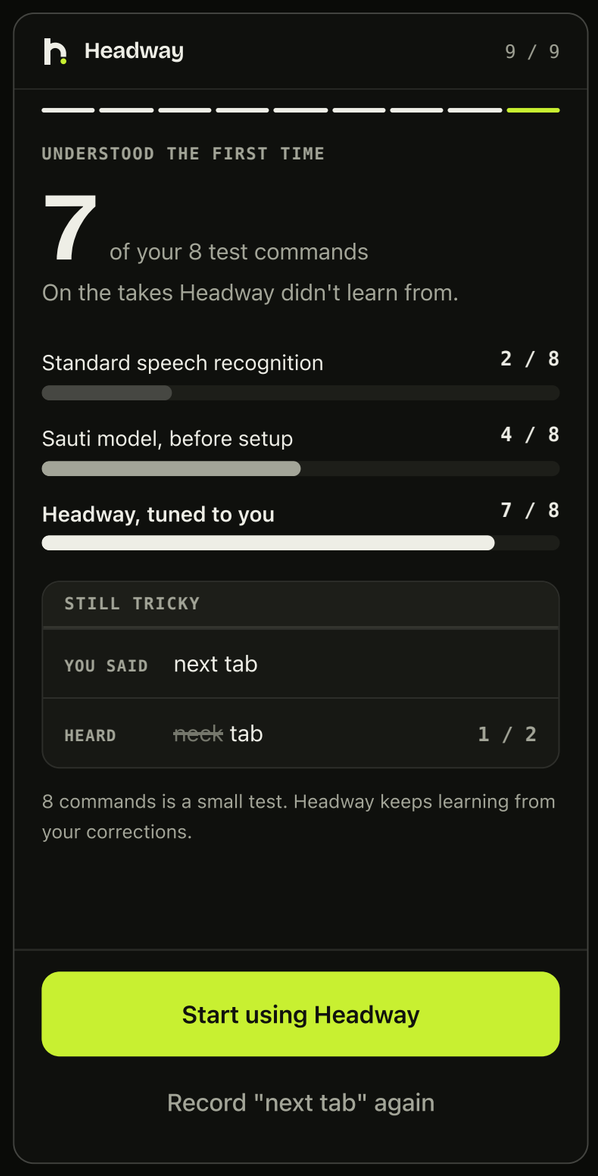
Fatigue is a design input. Breaks are offered, never forced; "Finish later" keeps every take and resumes at the next command.

09 · Answers and checks presentation



Her choice overrides inference. The toolkit's `presentation.js` turns confirmations on from a "motor" profile; Headway sets `confirmActions` from this answer instead.

10 · Your results the before / after

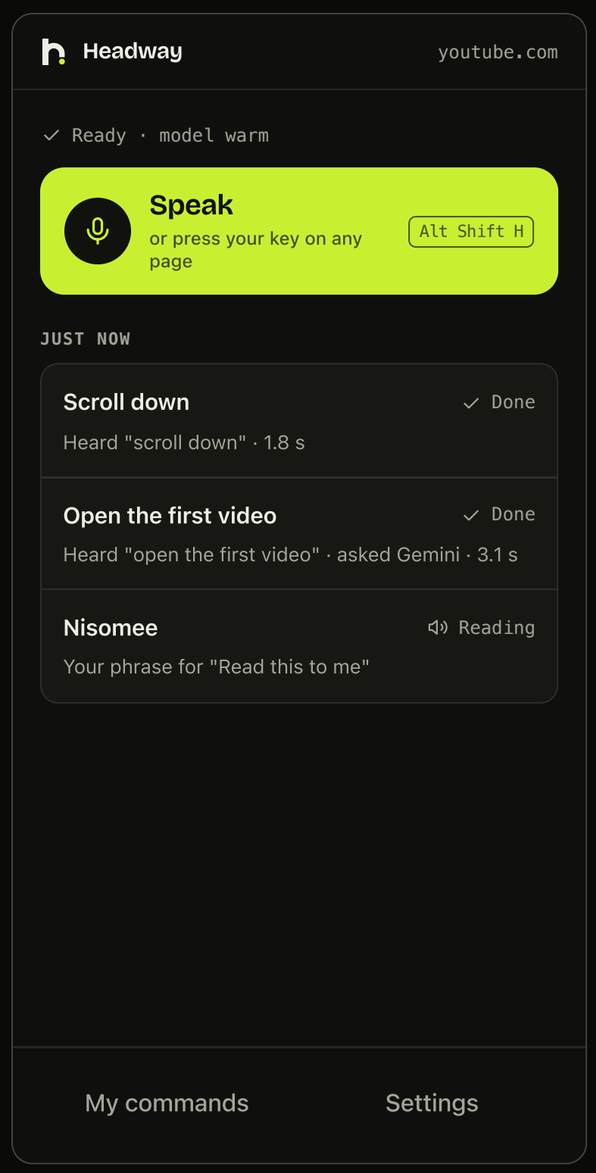


Honest numbers only. Computed live from her held-out takes, against both baselines. The sample size is on the screen. Values here are placeholders.

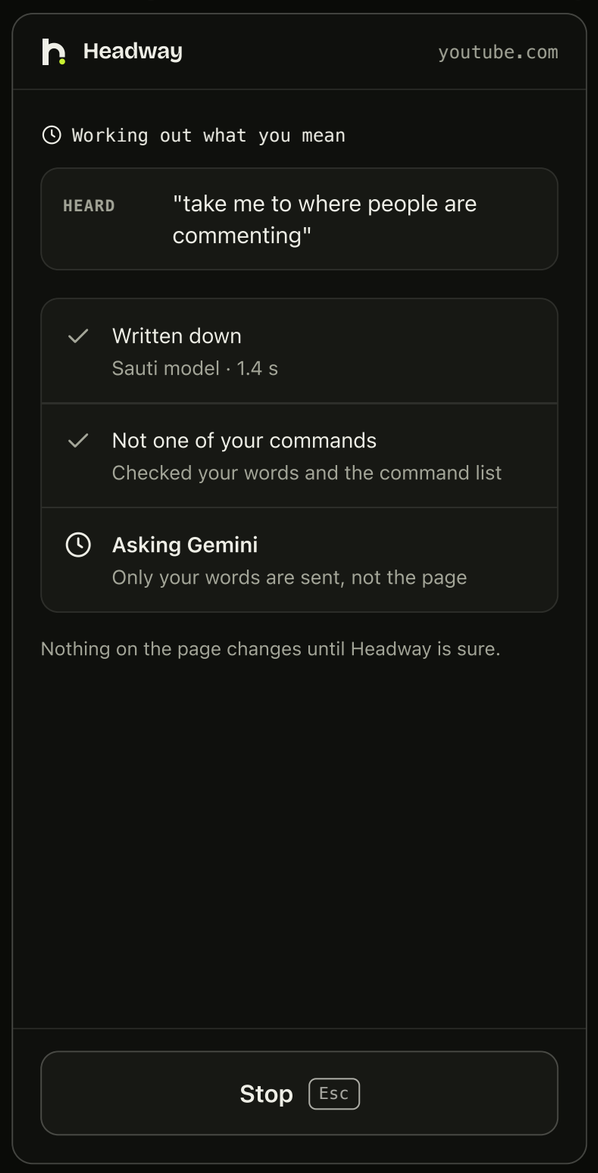
After setup: the panel in use

on any permitted page

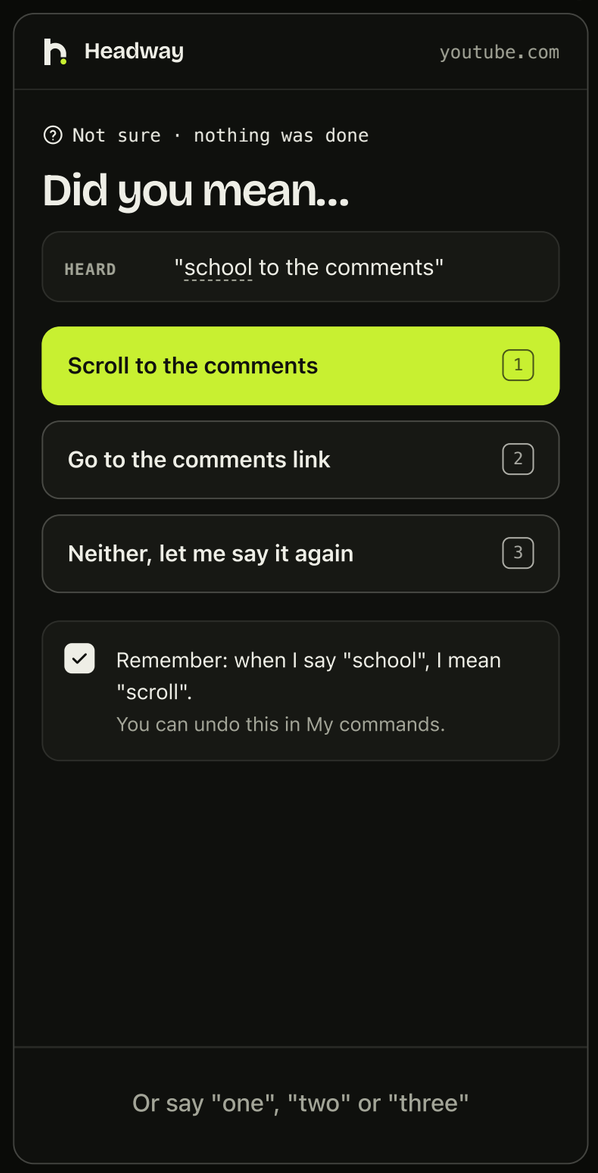
11 · Ready — home



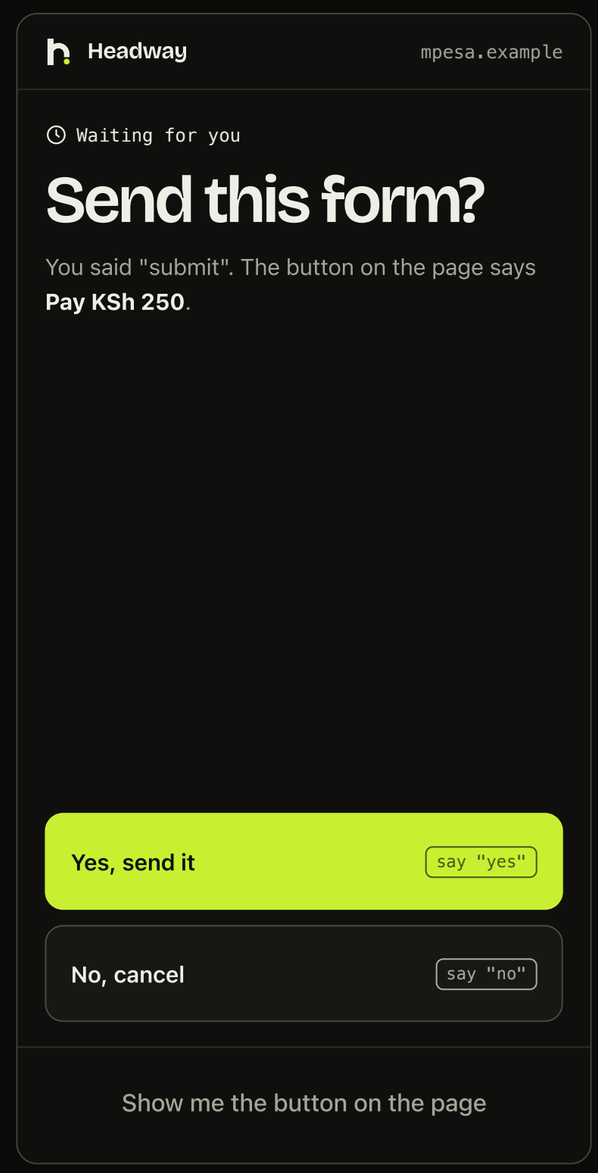
11b · Working it out — Gemini lane running



12 · Not sure — default to asking



13 · Check before sending — confirmation



Times are shown so the team can see latency in real use. "asked Gemini" marks the slow path; common commands never leave the command list.

The slow path is visible. She sees which step it's on, and Stop calls the Controller's `stop()`. Most commands never get here.

Learning by correction. The tick becomes a toolkit Librarian *proposal* she accepted, never a silent change. This is personalization that works on day one, before any model training.

Reads the real label back, not Gemini's summary of it, so a mis-heard "submit" can be caught before money moves. Uses the Controller's built-in yes/no pending state.